

## Lesson 4.13: Furnish It on a Budget

**Unit:** Unit 4, Spaces We Live In (Environmental Design and Management) **Topic:** 4.3 Design a Room project **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes (day 109) **Room:** classroom or FACS lab; calculators

### Standards

- NYS module line: NYS EDM 3. Consumer Resources and Finance a) Construct a budget for implementation of a floor plan design
- NYS module line: NYS EDM 3. Consumer Resources and Finance b) Use consumer skills to select household items
- CTE theme line: CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses
- CTE theme line: CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students are able to budget their time and money.
- CDOS Standard 3a Universal Foundation Skills: Managing Resources
- National FCS 3.0: 11.6.1 Work out what money and resources an interior improvement needs.
- National FCS 3.0: 2.1.5 Use consumer skills in choices about housing, utilities, and furnishings.

### Enduring understanding and essential question

**Big idea:** The plan is a wish until it is paid for. A budget forces the trade-off: new, secondhand, or make it, and which need gets the money first. **Essential question:** When the money runs out before the list does, what goes?

### Objectives

By the end of this lesson, students will be able to:

1. (Low) Given the furnishing catalog, find the new, secondhand, and DIY price of an item on the plan and record it with the item number.
2. (Mid) Price every piece on the plan, add sales tax at 8.625 percent, keep a running total, and bring the total at or under the client's budget (\$400 for a room refresh, \$1,500 for the tiny house interior).
3. (High) Justify three purchase decisions (one new, one secondhand, one DIY or free) by comparing cost and quality against the client's need, using the five-step decision model.

## Vocabulary

Tier 2 (general academic): budget, trade-off, priority, running total, justify Tier 3 (FACS): sales tax, subtotal, secondhand, DIY (do it yourself), unit price (from Lesson 1.16), quality versus cost. See Vocabulary 04.md for Spanish.

## Materials and setup

- Handout 04.13 - Furnishing Catalog and Budget Sheet.md: the catalog (40 items, three prices each) printed once per pair on two sheets, and the Budget Sheet, one per student or team
- Calculators, one per student (a calculator is allowed all period)
- The plan from Lesson 4.12 and the client card, in the project folder
- Highlighters, one per pair
- Slides: Slides 04.13 - Furnish It on a Budget
- Optional: two or three real listings printed from a local marketplace or thrift store site for the hook, with the seller's name and photo cropped out
- Setup notes: check the exit cards from Lesson 4.12 for path math fails and pull those students for a two-minute check during the do now. Confirm the sales tax rate for the school's county and write it on the board.

## Pre-assessment

The do now (a sketch from memory) shows how well students know their own plan. The first two minutes of Step 1 (a hand vote on “who has bought something secondhand, and what”) shows how much explaining secondhand needs.

## Do now and hook (Time: 5 min)

Floor plan do now (stack, sketch 8): “Sketch your client's room from memory. No looking. Two minutes. Then open your plan and compare. Write one thing you forgot.” One minute to compare. Sheets go in the stack.

Hook (slide 2): show the same item three ways. A 4-foot desk: new in a box, \$149; the same desk secondhand from a marketplace listing, \$40; a DIY version (a door blank on two file cabinets), \$35 in parts. “Same job. Three prices. Your client has \$400 for the whole room. Which desk, and what does that decision buy you somewhere else?”

## Procedure

| Step                                                          | Teacher will                                                                                                                            | Students will                                                                          | Time  |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------|
| 1. Direct instruction: the budget, the tax, the three columns | Slides 3 to 6. The budget comes from the client card: \$400 for a room refresh, \$1,500 for the tiny house interior. Say what a refresh | Copy the tax math line. Fill the header of the Budget Sheet: client, budget, tax rate. | 8 min |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                              |        |
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|                                      | <p>means: the walls and the floor are there; the money buys furniture, storage, light, paint, and finish. Show the catalog format: item number, item, size, new price, secondhand price, DIY price and what DIY means for that item, and a quality note. Show the tax: in New York, furnishings and building supplies are taxed; food is not (from Unit 1). The rate here is 8.625 percent. Subtotal times 1.08625 equals the total, or subtotal times 0.08625 equals the tax. Work one line on the board: twin bed frame, secondhand, \$60; tax \$5.18; total \$65.18. Say the story from business: “A budget is not a limit on your ideas. It is the order you buy them in.”</p> |                                                                                                                                                                              |        |
| 2. Learning activity: price the plan | <p>Slide 7. Release. Every piece of furniture on the plan is a line on the Budget Sheet: item number, which column (new, secondhand, DIY), price. Needs first,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Price every piece from the plan. Running subtotal after each line. Stop and re-decide when the subtotal passes 80 percent of the budget with needs still on the list.</p> | 11 min |

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|                              | <p>wants after. Keep a running subtotal. Circulate with the clipboard. Push on the column choice: “Why new for the mattress and secondhand for the dresser?” (The catalog’s quality note says mattresses and crib mattresses are not sold secondhand; safety and hygiene.)</p>                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                |        |
| 3. The tax and the trade-off | <p>Slide 8. Call time. Everyone computes the tax and the total. Over budget? The trade-off box: cross out a want, switch a column, or remove a piece from the plan (and the plan changes too). Under budget with money left? Add a want, or write the leftover as the client’s cushion. Model the five-step decision model on one team’s hardest choice: name the choice (armchair new at \$189 or secondhand at \$45), list options, weigh each against what matters to the client (Grandpa needs arms and a firm seat; a secondhand chair must be checked for that), decide, look back.</p> | <p>Compute tax and total. Revise until the total is at or under budget. Write three justifications on the sheet: one new, one secondhand, one DIY or free, each with the reason tied to the client’s need.</p> | 10 min |

|          |                                                                                                                                                                                                                                            |                                                |       |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------|
| 4. Share | Slide 9. Two teams report: the total, the biggest single line, and the hardest trade-off. Class question: was the trade-off a want or a need? Preview Lesson 4.14: the model is built from scrap because that is the DIY column made real. | Report or listen. Sign the Budget Sheet total. | 3 min |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------|

### Questions to ask

Monitor understanding:

- What is the subtotal, the tax, and the total on your sheet right now? Show me the line where you multiplied.
- Which item did you buy new, and what is the reason on the sheet?
- What does the quality note say about secondhand mattresses?

Deepen learning:

- A secondhand dresser at \$30 has a broken drawer. The new one is \$179. What does “cheaper” actually cost here? (Time, a repair, or a drawer that never works.)
- The tiny house couple has \$1,500 and a kitchen to furnish. What would you buy first and why? What could wait until they move in?
- Your family bought something for your home this year. Do you know if it was new or secondhand? Would you have chosen differently?

### Check for understanding

Clipboard check during Step 2: for each student or team, a yes or no on “every plan piece has a line,” “running subtotal is correct at the line I checked,” and “column choice has a reason when asked.” A team with a wrong running subtotal recalculates from line 1 with a partner before Step 3, the same recheck routine as Lesson 1.19.

### Closure (Time: 3 min)

Exit card: “My total is \$\_\_ against a budget of \$. **The hardest trade-off was** and I chose \_\_ because \_\_.” Two students read theirs. Sort by over or under budget; anyone over gets the first five minutes of Lesson 4.14 Day 1 to fix it before building.

## Differentiation and supports

- ELL: the catalog has a small picture line for each item; the three columns are labeled with icons (a box for new, a recycle arrow for secondhand, a hammer for DIY); the Budget Sheet has the justification frame "I chose \_\_\_ (new / secondhand / DIY) for \_\_\_ because the client needs \_\_\_"; the words budget, tax, total, new, secondhand, and make in Spanish on the sheet margin (see Vocabulary 04.md).
- IEP and 504 (general): a calculator with a tax key or a printed tax table (every \$10 from \$10 to \$400 with the tax already computed) on the back of the Budget Sheet; a shortened plan of five required pieces for a student who needs it; a partner runs the calculator while the student chooses columns; extended time into Lesson 4.14 Day 1.
- Grade 6 support: price only the needs, use the tax table instead of multiplying, and write one justification instead of three.
- Grade 8 stretch: compute the percentage of the budget spent on each zone (sleep, work, store, light), and write a two-sentence recommendation to the client on what to buy this month and what to wait on, the way a designer would present a phased budget.
- UDL checkpoint used: multiple means of action and expression (calculator, tax table, or long multiplication all accepted; justification written or dictated to a partner) and multiple means of engagement (the client's money is the constraint, not the teacher's rule).
- No-kitchen or no-machine alternative: not a lab; no alternative needed.

## Assessment

| Objective                                                           | Standards met                                         | Learning task                     | Assessment (formative or summative; tool)                                                          |
|---------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------|
| 1. Find and record the three prices                                 | EDM 3. b); CDOS 3a Managing Resources                 | Step 2, Budget Sheet lines        | Formative; clipboard check                                                                         |
| 2. Price the plan, add tax, keep a running total at or under budget | EDM 3. a); FCL 4. c); Std 3 PI C; National FCS 11.6.1 | Steps 2 and 3, Budget Sheet total | Formative today (exit card); summative on Rubric 04 criterion 3 and on the Unit 4 Test budget item |
| 3. Justify three purchase decisions                                 | EDM 3. b); FCL 4. f); National FCS 2.1.5              | Step 3, justification box         | Formative; the three justifications are read for Rubric 04 criterion 3                             |

## Homework

None. Optional: look up one item from your Budget Sheet on a real store site or a marketplace listing (with an adult). Write the real price next to the catalog price. Was the catalog close?

## Connections

Math: percent, multiplying by a decimal, running totals; this is the same tax math the math teacher can use for a percent unit. Business or economics, if the school has it: new versus used markets and why a secondhand price is set by the seller. ClassroomStreet: the furnishing total can be entered as a one-time expense in a student's ClassroomStreet budget in Unit 2's follow-up run, so a room refresh shows up against a real monthly income.

## Sources

- NYS Middle Level CTE FACS Content Module 04, Environmental Design and Management (nyc-tecenter.org, June 2018), lines 3. a) and 3. b). Copy in 02 Standards/NYS Module PDFs/. The module's Financial and Consumer Literacy illustrative activity (renting versus buying furniture) is the source of the "what does cheaper cost" question.
- NYS Middle Level CTE Theme Module, Financial and Consumer Literacy, lines 4. c) and 4. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS SOURCES - web research.md, section 2.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 11.6.1 and 2.1.5, per 02 Standards/STANDARDS SOURCES - web research.md, sections 4.2 and 4.11.
- Sales tax: New York State Department of Taxation and Finance, Publication 718 (sales and use tax rates by jurisdiction).
- My own materials: the \$40 grocery project's pricing routine (Unit 1, Lessons 1.18 and 1.19) and the "budget" vocabulary entry from. Catalog prices are realistic 2026 figures for Long Island and are every one to be refreshed before teaching.

## Teacher notes

- If the period runs short, cut Step 4 and take the two reports at the start of Lesson 4.14. Never cut the tax line; the Unit 4 Test has a tax item.
- Common mistake: adding the tax to every line instead of once at the bottom. Both give the same total, but the line-by-line version doubles the chance of an error. Teach it as subtotal first, tax once.
- Common mistake: the DIY column read as free. DIY has a parts price; the labor is the client's. Ask "who builds it, and do they know how?"
- The mattress rule (no secondhand mattresses or crib mattresses) is on the catalog. It is a hygiene and safety point, and for the crib it connects to safe sleep from Lesson 3.17.
- Keep the Budget Sheets in the project folder. Criterion 3 is scored from the sheet, not from memory.